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Four decades of spatio-temporal habitat dynamics in a human-wildlife co-existence landscape in the Liuwa ecosystem of Zambia

Danny Chisanga Musenge¹, Darius Phiri^{2*}, Ngawo Namukonde³, Donald Zulu², Gift Mulenga¹ and Vincent R. Nyirenda³

*Correspondence:

Darius Phiri

dariusphiri@rocketmail.com

¹School of Graduate Studies, Copperbelt University, Off Kalembe Driver, Parklands, P.O. Box 21692, Kitwe, Zambia

²Department of Plant and Environmental Sciences, School of Natural Resources, Copperbelt University, Jambo Drive, Riverside, P.O. Box 21692, Kitwe, Zambia

³Department of Zoology and Aquatic Sciences, School of Natural Resources, Copperbelt University, P.O. Box 21692, Kitwe, Zambia

Abstract

Human encroachment into wildlife habitats has led to extensive habitat loss and fragmentation, often resulting in the displacement and decline of wildlife populations across African Ecosystem. Meanwhile, Habitat is operationally defined as natural land cover types that directly support wildlife, namely forest, grassland, and wetland, therefore, this study assessed four decades (1984—2024) of spatio-temporal habitat dynamics in Liuwa Plain National Park (NP), a human-wildlife co-existence landscape of Zambia. Habitat trajectories inside and outside the park were compared to evaluate the protective effect of co-management. Using Landsat satellite imagery classified at 10-year intervals (1984, 1994, 2004, 2014, and 2024) and a random forest algorithm with a two-step approach: pixel-based classification followed by an object-based refinement, we produced thematic maps with accuracies of 82—89%, with the 2024 map being the most accurate. This study assessed habitat extent rather than habitat quality. Results show a 20% decline in forest cover, a 5% loss of grasslands, and a 3% reduction in wetlands and water bodies in 2024, likely due to the 2023/2024 drought. Habitat loss was greater outside the park than inside, concentrated in the southern Kalabo Urban Zone and more evident outside the park boundaries. These shifts indicate rising anthropogenic pressure on habitat critical to wildlife persistence. The study underscores the need for integrated land-use planning, habitat management, adaptive park zonation and supports the identification of ecological hotspots to sustain co-existence and ecological integrity in Liuwa NP and similar landscapes.

Keywords Biodiversity conservation, Blue-wildebeest, Human-wildlife interaction, Land cover changes, Landsat

1 Introduction

Landscapes in protected areas such as National Parks (NPs) change because of natural disruptions and anthropogenic activities; however, anthropogenic activities are by far the most significant threat to biodiversity in protected areas [1]. Many scholars have highlighted how human encroachment into wildlife habitats has led to widespread habitat



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loss and fragmentation, often resulting in the displacement and decline of wildlife populations [2, 3]. Traditionally, the establishment of protected areas such as game reserves and national parks has involved the relocation of rural communities, frequently implemented with limited consultation or inadequate compensation [4, 5]. In contrast, Liuwa Plain NP represents a unique conservation model where human and wildlife populations coexist within the landscape [6, 7]. This approach challenges conventional conservation paradigms and offers a valuable opportunity to examine alternative strategies to biodiversity conservation, while potentially reducing human-wildlife conflicts [8].

Unlike conventional protected areas where human habitation is restricted, Liuwa Plain NP is formally co-managed, with communities and wildlife sharing the same space [6, 7]. This distinguishes it from surrounding Game Management Areas (GMAs), where wildlife protection is secondary to utilisation, and from exclusionary parks where community access is prohibited. Understanding habitat dynamics in such landscapes requires comparing trajectories inside and outside the park to evaluate whether co-management governance effectively mitigates habitat change.

Wildlife-based land use planning within protected areas, such as national parks, has significant potential to support community livelihoods through sustainable tourism and the provision of ecosystem services [9]. However, in regions like the Liuwa ecosystem, consisting of both the national park and its surrounding areas, where agricultural potential is low, the effectiveness of protected areas in delivering economic benefits from wildlife-based enterprises is often limited by natural resource abundance [6, 7]. As a result, many protected area networks are falling short of their full potential to contribute to ecological conservation, economic development, and social well-being [8]. This highlights the urgent need for effective management strategies and sustainable financing mechanisms [6]. Therefore, understanding and monitoring of long-term spatio-temporal habitat dynamics in the Liuwa Plain NP and the surrounding areas will provide crucial information for management regimes and restoration of stressed land cover types within the ecosystem and provide conducive habitats for wildlife and other secondary dependents such as humans, as well as establishing informed conservation priorities. In Zambia, where issues such as encroachment and illegal wildlife trade remain prevalent, there is a pressing need for comprehensive data on habitat dynamics in terms of spatio-temporal extent to guide conservation practice and policy decisions.

Numerous studies have explored various aspects of resource management in Zambia's protected areas, addressing themes such as illegal wildlife harvesting [10, 11], land encroachment and human-wildlife conflicts [12, 13], policy frameworks [8], ecological dynamics [14, 15] and community participation [16, 17], however, few have combined robust spatio-temporal habitat modelling with the lens of a shared landscape [7]. Prior studies in Zambian and Southern African protected areas, including Kpienbaareh et al. [1] in Kasungu NP and Banda et al. [18] in the Barotse Floodplain, focused on exclusionary parks or examined wetland change without accounting for co-managed, human-inhabited landscapes. Similarly, assessments in Gonarezhou NP [19] and the Okavango Delta [20] documented land cover change but did not systematically compare inside-versus-outside park trajectories over multi-decadal timescales. To our knowledge, no study has conducted a 40-year, inside-versus-outside comparative analysis of habitat dynamics in an African co-managed protected area. Advances in remote sensing, spatio modelling and change detection offer new opportunities to fill the gap, for instance, recent work by

[21] utilised geospatial and machine learning approaches to parse long-term land cover changes and urbanisation patterns in other contexts, for example, the spatio-temporal expansion in Lucknow, India and to model complex environmental changes under combined anthropogenic and natural pressures as employed in the study by [22]. As such, the studies highlighted above indicate the feasibility and power to integrate spatio-temporal and analytical remote sensing techniques to untangle landscape dynamics.

The overall objective of this study is to assess four decades of spatio-temporal habitat dynamics in Liuwa Plain NP and its surrounding areas, focusing on forests, settlements, wetlands, grasslands, water, cropland and bareland. Further, the study aimed to achieve specific objectives to understand long-term land cover changes within the Liuwa ecosystem. Specifically, it sought to: (1) examine how land cover types have changed over the four decade from 1984 –2024 through the development of change maps for each decade; (2) compare habitat dynamics inside Liuwa Plain NP and the surrounding 40-km buffer; (3) investigate the patterns of habitat loss and gains across the ecosystem in order to determine the extent and spatial distribution of these changes, lastly; and (4) analyse the habitat transition matrices to illustrate the magnitude and direction of the land cover conversions highlighting how land cover classes have been transformed over time within the ecosystem. For the purposes of this study, habitat is operationally defined as the natural land cover types that directly support wildlife persistence, specifically forest, grassland, and wetland. Cropland, settlements, bareland, and open water are classified as non-habitat land cover. We hypothesise that: (i) natural habitat cover has declined across the Liuwa ecosystem over 40 years; (ii) habitat loss is greater outside the park than inside, reflecting the protective effect of co-management; and (iii) non-habitat land cover has expanded disproportionately in the buffer zone near Kalabo town. We note that this study assesses habitat extent (spatial area) rather than habitat quality (vegetation condition or ecological integrity). The findings are expected to offer new insights into the sustainable management of wildlife resources in settings where human communities and natural ecosystems share spaces.

2 Methods

2.1 Study area

Liuwa Plain NP, situated at latitude of -14.64 and a longitude of 22.74 (Fig. 1), is a unique conservation area in Zambia where wildlife co-exists with humans [7]. The total study area, encompassing Liuwa Plain NP ($3,660 \text{ km}^2$). Further, the study area comprises two spatial units: (1) Liuwa Plain National Park, and (2) a 40-km Upper West Zambezi Game Management Area buffer zone surrounding the park. Meanwhile the buffer zone covers approximately $16,966 \text{ km}^2$. additionally, a 40-km buffer was selected to capture the principal zone of anthropogenic influence around the park, encompassing the area where land-use transitions associated with Kalabo town occur. This distance also approximates the seasonal movement range of blue wildebeest in the ecosystem. Analysis was conducted within a 40 km buffer surrounding the study area. The Liuwa Plain NP was established in 1972, the park's conservation legacy dates to the nineteenth century when it was managed as a royal hunting ground under King Lewanika of the Lozi People. The park is characterised by a floodplain with an average elevation of 1050 m above mean sea level [23]. The park is unfenced, allowing wildlife and people to move freely across boundaries. The area is inhabited by local communities engaged in small-scale

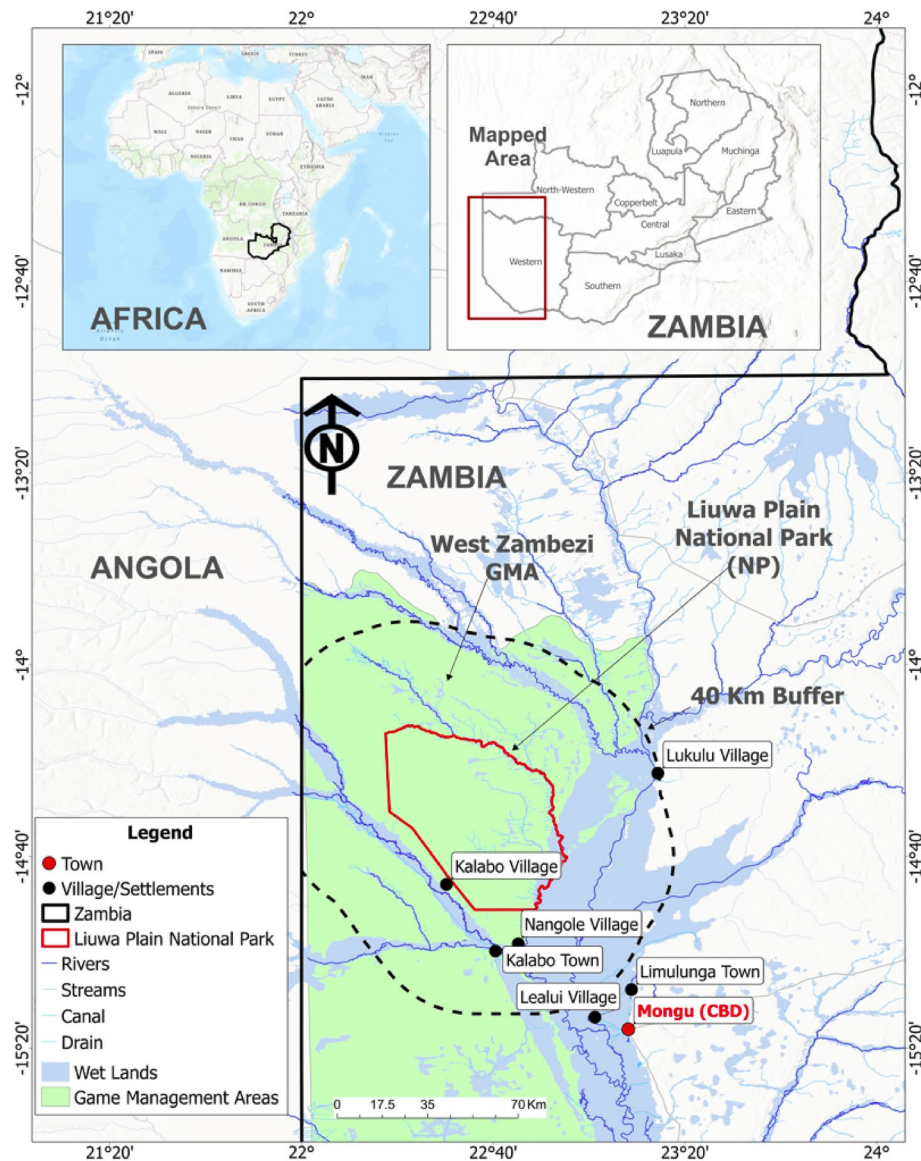


Fig. 1 The study area includes Liuwa Plain National Park and the surrounding 40 km Upper West Zambezi GMA buffer zone, that includes and the four villages/settlements (Kalabo village, Kalabo town, Nangole Village, and Lukulu village)

mixed-farming, fishing, and other natural resource-based enterprises. Since 2003, the park has been jointly managed by the Department of National Parks and Wildlife (DNPW), African Parks Network (APN), and the Barotse Royal Establishment (BRE), promoting effective natural resource management and community participation [17]. Besides, approximately over 10,000 community members reside within the park, creating a rare functioning co-existence landscape [6, 7], although there is no formal residential permit system within Liuwa Plain NP, community habitation is regulated through the tripartite co-management agreement which establishes land-use zones delineating areas for community settlement, agriculture, and core wildlife conservation. New settlements require approval from the BRE in consultation with park management [24].

The park supports the second-largest wildebeest migration in Africa, with over 40,000 blue wildebeest (*Connochaetes taurinus*) [25], globally significant wattled crane

(*Bugeranus carunculatus*) populations and over 300 bird species, including endangered wattled cranes, grey crowned cranes, and saddlebill storks. Large mammals inhabiting the park include blue wildebeest (*Connochaetes taurinus*), Burchell's zebra (*Equus quagga*), and tsessebe (*Damaliscus lunatus*), with a large annual locally migratory blue wildebeest (*Connochaetes taurinus*) population, second to Serengeti in East Africa [25]. The park also hosts predators such as lions (*Panthera Leo*), cheetahs (*Acinonyx jubatus*), and spotted hyenas (*Crocuta crocuta*), making it a significant conservation area in Zambia.

The Liuwa ecosystem is situated within Agro-Ecological Regional IIB [24], which is dominated by sandy soils and receives mean annual rainfall of 800–1,000 mm with a unimodal wet season (November–April) and dry season (May–October). Mean annual temperatures ranging from 8.1–34.9 °C. The park's ecosystem is dominated by upland grasslands and seasonally flooded grasslands which transition to wetland conditions during the wet season and most part of the year, with standing water typically persisting from November through most months of the concurrent year, supporting diverse biodiversity [7]. The study area is characterised by Kalahari woodlands, which consist of a mosaic of dry deciduous woodland vegetation adapted to deep Kalahari sand substrates, are dominated by species including Zambezi teak (*Cryptosepalum exfoliatum*), Rhodesian teak (*Baikiaea plurijuga*), and *Brachystegia* spp, while grasslands are characterised by common russet grass (*Loudetia simplex*) and boat grass (*Monocymbium cerasiiforme*). followed by the list of woodland specieswoodland.

2.2 Materials and methods

2.2.1 Overall workflow

Figure 2 shows the workflow used to achieve the objectives of this study. The methodological approach followed a systematic workflow, ensuring clear and logical progression from data acquisition to analysis. The study integrated remote sensing, GIS, and analytical techniques to assess habitat dynamics over 40 years (1984–2024) in Liuwa Plain NP, a human-wildlife co-existence landscape of Zambia. The methodology is divided into four

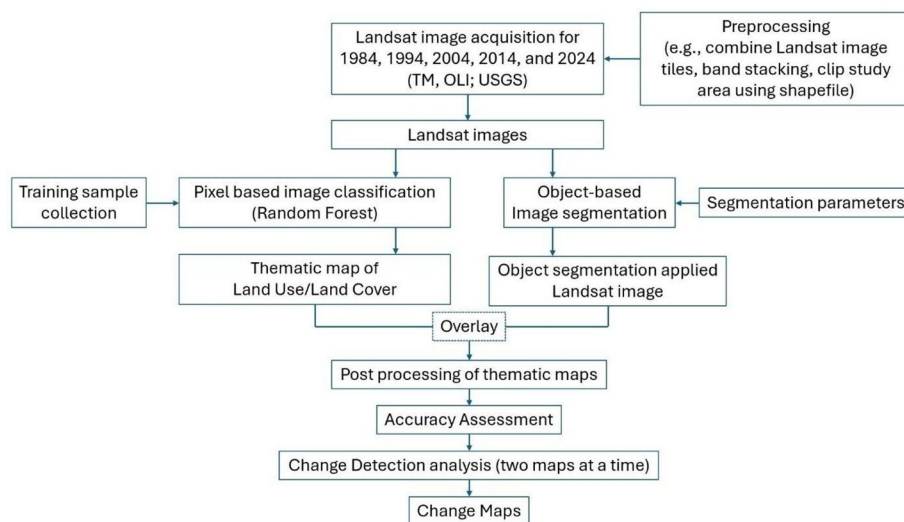


Fig. 2 The main data processing steps used for each year's image include data acquisition, the two-step image classification, accuracy assessment, and change detection (See a detailed flow chart under the Supplementary Material)

key phases: (1) data acquisition and pre-processing, (2) LULC classification using a combination of pixel-based classification (PBC) and object-based post-classification refinement (OBPR), (3) Change detection and analysis to examine spatial trends in landscape transformation both within and outside park boundaries; and (4) Results interpretation.

2.2.2 Data acquisition and pre-processing

2.2.2.1 Imagery dataset This study employed multi-temporal Landsat imagery (1984–2024) to derive land-use and land cover (LULC) maps. Landsat 5 TM images (1984, 1994, 2004, and 2014) and Landsat 8 OLI/TIRS imagery (2024) were obtained from the USGS Earth Explorer platform. Collection 2 Level 2 surface reflectance which are products, atmospherically corrected using LaSRC (Landsat8) and LEDAPS (Landsat 5), were used, eliminating the need for additional atmospheric pre-processing while the Landsat 5 TM bands 1 (Blue), 2 (Green), 3 (Red), 4 (Near Infrared), 5 (Shortwave Infrared 1), and 7 (Shortwave Infrared 2), and Landsat 8 OLI bands 2 (Blue), 3 (Green), 4 (Red), 5 (Near Infrared), 6 (Shortwave Infrared 1), and 7 (Shortwave Infrared 2) were used for each year's Landsat image, individual spectral bands were stacked into a multi-band composite image. The two adjacent Landsat tiles (Path/Row 172/068 and 172/069) were then mosaicked to produce a seamless image covering the full study area extent. The Analysis was conducted both within Liuwa Plain NP and the 40 km Upper West Zambezi GMA outside the park. was subsequently used to clip the mosaicked composite, yielding analysis-ready imagery for classification. Moreover, for each year, images with less than 10% cloud cover were downloaded, and for each year's Landsat image, all the bands with a 30 m spatial resolution were selected to create a stacked image. No derived layers (e.g., NDVI, DEM) were included in the classification stack. All images were acquired during the late dry season (August–September), when vegetation phenology is most distinct. Single-date imagery per time point was used due to limited cloud-free availability; consistent dry-season acquisition minimises phenological differences between periods [26, 27]. The imagery was projected to UTM Zone 35S (EPSG:32,735). While Landsat 5 TM and Landsat 8 OLI differ in spectral bandwidth, spectral response functions, and radiometric resolution, both are processed to surface reflectance, which substantially improves cross-sensor comparability by reducing atmospheric effects [26]. However, surface reflectance processing does not fully eliminate these inherent sensor differences. Residual spectral differences may introduce systematic biases between the 2014/2024 (Landsat 8) and earlier (Landsat 5) classifications [28] (Table 1).

2.2.2.2 Auxiliary dataset Auxiliary spatial datasets, including boundary shapefiles which were obtained from the National Remote Sensing Centre (NRSC) in 2024, topographic maps were equally sourced from the NRSC (1:50,000 series, 2005). Digital elevation models (DEMs), topographic maps, and Google Earth imagery from 2022 was utilized to support training data collection and validation and Settlement extent data which were obtained from the GRID3 Zambia Settlement Extents v3.0 dataset (<https://data.grid3.org/datasets/GRID3::grid3-zmb-settlement-extents-v3-0/about>). The boundary

Table 1 Details of the images used in this study include the type of Landsat images, date of acquisition, spatial resolution, and satellite path/row

Sensor	Date of Acquisition	Spatial resolution (m)	Path/Row
Landsat 5 TM	1984/09/13	30	172/068, 172/069
Landsat 5 TM	1994/08/29	30	172/068, 172/069
Landsat 5 TM	2004/08/14	30	172/068, 172/069
Landsat 8 OLI	2014/09/08	30	172/068, 172/069
Landsat 8 OLI	2024/09/12	30	172/068, 172/069

TM Thematic Mapper, OLI Operation Land Imager, USGS United States Geological Survey

shapefile delineated the park and buffer zones. Topographic maps and the DEM served as visual aids during training data collection; the DEM was not stacked with Landsat bands as a classification input. In addition, the Digital Elevation Models (DEMs) were acquired from the Shuttle Radar Topography Mission (SRTM) 30-m product provided by USGS (acquired in 2000). These complementary datasets were integrated to enhance the accuracy and interpretation of habitat dynamics within the study area.

2.2.3 LULC classification

2.2.3.1 Classification scheme development The LULC classification was conducted using a scheme comprising seven distinct classes: Settlements, Forest, Cropland, Wetlands, Grassland, Bareland, and Water (Table 2). The development of these classes was informed by reconnaissance surveys and expert knowledge of the study area, ensuring relevance and accuracy. This classification system allowed for a nuanced understanding of land use patterns and changes within the study area.

2.2.3.2 Image classification This study implemented a two-step approach combining pixel-based classification (PBC) and object-based post-classification refinement (OBPR). The accuracy and reliability of this integrated classification methodology for LULC mapping have been established in earlier studies [24, 26, 29].

2.2.3.3 Pixel-based classification Training data for each time point were digitised on-screen as polygon regions of interest (ROIs) over the respective Landsat composite by a single analyst (DCM) to ensure consistency. Region of Interests (ROIs) were delineated alongside the ESRI imagery hybrid basemap, Google Earth imagery, and boundary shapefiles for spatial reference. The pixel-based classification was performed in ArcGIS Pro 3.5.1 using the Image Classification Wizard, with a Random Forest classifier (100 trees, maximum depth 30, 2,000 samples per class). Random Forest was selected for its demonstrated superiority in heterogeneous African landscapes [26, 29], robustness to noise, and ability to handle moderate-dimensional feature spaces without extensive tuning. A maximum of 100 trees, a maximum tree depth of 30, and a maximum of 2,000 samples per class. [26, 29].

2.2.3.4 Object-based post-classification refinement Following pixel-based classification, OBPR was implemented to address misclassifications arising from spectral confu-

Table 2 Land use/cover scheme. Forest, grassland, and wetland are the most ecologically significant classes for wildlife habitat

Land use/cover classes	Description
Forest	Deciduous forests, evergreen forests, mixed forest lands, coniferous forests, orchards, commercial plantations, protected areas, and nurseries
Settlements	Urban/rural settlements with transportation and communication facilities
Wetland	Areas characterised by water and close to water sources, such as rivers, major grass species are; Hippo grass (<i>Echinochloa stagnina</i>), Water grass (<i>Vossia cuspidate</i>), Golden grass (<i>Loudetia simplex</i>) and oat grass (<i>Monocymbium cerasiiforme</i>)
Grasslands	Stunted height degraded forest, shrubs, or grasses
Water	Rivers, lakes, ponds, reservoirs/dams, streams
Cropland	Pastures, orchards, home gardens, and areas covered with perennial and annual crops, may be fallow
Bare lands	Barren areas devoid of vegetation cover and consisting of exposed soils and rock outcrops or sandy surfaces

sion between land cover classes [29]. The two-step approach was adopted because OBPR leverages spatial context, shape, and neighbourhood relationships to correct misclassifications while preserving ecologically meaningful small patches unlike a simple majority filter, which may eliminate such features [26, 30, 31].

Using eCognition Developer software (version 10.3), multi-resolution segmentation was applied to each year's Landsat composite to partition the imagery into homogeneous image objects for each time point [27]. Fixed parameters of scale 13, shape 0.2, and compactness 0.8 were used; All five stacked Landsat images were segmented with the same set of segmentation parameters. The Liuwa floodplain presents challenges for class separability due to spectral overlap between wet grasslands, wetlands, and shallow water; the OBPR was designed to address these challenges. Pixels were identified using a systematic rule-based overlay protocol: the pixel-based Random Forest classification was overlaid onto the segmented image objects, and for each object, the majority land cover class was computed. Where the majority class disagreed with more than 30% of constituent pixels, the object was flagged for review by DCM against ancillary spatial data (ESRI imagery hybrid basemap, Google Earth high spatial resolution imagery, and boundary shapefiles) [32]. Flagged objects were reassigned a single LULC label based on majority-rule voting within the object boundary. The corrected labels were manually assigned to the 30-m pixel grid. Approximately 8–15% of map pixels were edited per classification year. The corrected outputs were integrated into the original maps, producing five refined LULC classifications for 1984, 1994, 2004, 2014, and 2024 [33]. Mosaicking may introduce minor resampling artefacts, but the impact is expected to be minimal given the small proportion of reclassified segments. To address the potential circularity between OBPR and accuracy assessment, validation points were generated independently of the refinement process using stratified random sampling (Sect. 2.2.4), and no validation point coincided with segments that had been manually reclassified during the OBPR step.

2.2.4 Accuracy assessment

Accuracy was assessed using 300 stratified-random validation points, generated in ArcGIS Pro using the Create Accuracy Assessment Points tool (≥ 40 per class) per time point. The reference labels generated by ArcGIS Pro were inspected by a separate analyst, who visually compared each pixel against high spatial resolution Google Earth imagery, and data from original Landsat images and field based ground truthing [27, 34]. The error matrix (confusion matrix) was generated using the Compute Confusion Matrix tool in ArcGIS Pro, from which the following metrics were computed:

Overall accuracy (OA):

$$OA = \frac{\sum_{i=1} x_{ii}}{n} \times 100 \quad (1)$$

Producer's accuracy (PA):

$$PA_i = \frac{x_{ii}}{x_i +} \times 100 \quad (2)$$

User's accuracy (UA)

$$UA_i = \frac{x_{ii}}{x_{+i}} \times 100 \quad (3)$$

where:

- x_{ii} = number of correctly classified pixels for class i (the diagonal element)
- x_{i+} = total reference pixels for class i (row sum)
- x_{+i} = total classified pixels for class i (column sum)
- n = total number of reference points
- r = total number of classes

2.2.5 Change detection analysis

Post-classification change detection analysis was performed across five time intervals for two time periods at a time: 1984–1994, 1994–2004, 2004–2014, 2014–2024, and the full period 1984–2024. The post-classification refinement applied 'random forest' classified maps were used for comparison because it provides from to transition information essential for constructing transition matrices [27]. The ArcGIS Pro change detection workflow was utilised to analyse the change trajectories and determine the rate of change for each time step (See Eq. 4–6). In this study, change was defined as the net loss of natural habitat classes (forest, grassland, and wetland combined), calculated as the absolute difference in area (km²) between the initial and final time steps for each comparison period. The rate of change was calculated using the formulas in Microsoft Excel:

$$\text{AnnualRateofchange} = \frac{A_2 - A_1}{A_1 \times t} \times 100 \quad (4)$$

$$\Delta = A_2 - A_1 \quad (5)$$

$$\% \Delta A = \frac{A_2 - A_1}{A_1} \times 100 \quad (6)$$

where:

- A_1 = Area of a given LULC class at the beginning of the period
- A_2 = Area of the same class at the end of the period
- t = Length of the time interval in years

2.2.6 Spatial analysis and comparison

A separate analysis was conducted inside and outside the Liuwa Plain NP to evaluate conservation effectiveness (CEI) using Microsoft Excel (Eq. 7).

$$CEI = \frac{\text{Change}_{\text{outside}} - \text{Change}_{\text{inside}}}{\text{Change}_{\text{outside}}} \times 100 \quad (7)$$

where:

- $\text{Change}_{\text{inside}}$ = loss of natural habitat (forest, grassland, and wetland combined, in km²) within park boundaries
- $\text{Change}_{\text{outside}}$ = loss of natural habitat (forest, grassland, and wetland combined, in km²) in the 40-km buffer zone. Positive CEI values indicate that the park experienced less habitat loss than the surrounding buffer

2.2.7 Habitat transition matrix analysis

- Transition matrices identified donor (loss) and beneficiary (gain) classes. Matrices report gross transitions; net changes are derived by subtracting losses from gains. (Formulas 8, 9 and 10).

$$T_{ij} = \text{areatransitioningfromclass}i\text{at}time1\text{to}classj\text{at}time2$$

Then :

$$Gain_j = \sum_{i \neq 1} T_{ij} \quad (8)$$

$$Loss_i = \sum_{j \neq 1} T_{ij} \quad (9)$$

$$Persistence_j = \frac{T_{ii}}{A_{i(t1)}} \times 100 \quad (10)$$

where:

- T_{ij} = area transitioning from LULC class i to class j
- $Gain_j$ = total area gained by class j from all other classes
- $Loss_i$ = total area lost by class i to all other classes
- T_{ii} = area persisting in class i (diagonal element)
- $A_{i(t1)}$ = area of class i at the initial time step

3 Results

3.1 Land use/cover classification

The land cover maps generated for the study area demonstrated overall classification accuracies ranging from 82 to 89% (Table 3), indicating generally acceptable classification performance. The lowest overall accuracy was recorded for the 1984 map (82%), while the 2024 classification achieved the highest accuracy (89%). Users' and producers' accuracies ranged from 39 to 100%, with most classes above 80%. However, class-level performance was uneven. The 2024 Settlement class exhibited a low user's accuracy (39%), reflecting spectral confusion between small rural settlements and bareland at 30-m resolution. The 2024 Bareland user's accuracy (70%) was similarly affected. These class-specific limitations should be considered when interpreting fine-scale transitions

Table 3 Classification accuracy before applying the object-based refinery approach. The improvement ranged from 10–15%

Land cover	1984		1994		2004		2014		2024	
	PA (%)	UA (%)	PA (%)	UA (%)	PA (%)	UA (%)	PA (%)	UA (%)	PA (%)	UA (%)
Bareland	91	89	87	89	97	97	92	85	91	71
Cropland	87	81	88	88	98	98	83	91	77	79
Water	91	71	93	94	100	100	91	88	98	85
Grassland	88	83	78	88	79	94	88	91	80	88
Settlement	81	86	87	84	89	89	84	81	80	78
Forest	79	80	76	85	85	87	86	81	81	74
Wetland	91	89	88	91	98	98	81	86	88	77
OA (%)	72		74		79		80		81	

Table 4 Accuracy assessment of the final thematic maps which reports the user, producer and overall accuracies

Land cover	1984		1994		2004		2014		2024	
	PA (%)	UA (%)	PA (%)	UA (%)	PA (%)	UA (%)	PA (%)	UA (%)	PA (%)	UA (%)
Bareland	97	99	97	97	97	97	94	95	93	70
Cropland	92	86	98	98	98	98	93	93	78	82
Water	100	81	100	100	100	100	97	100	100	95
Grassland	100	93	79	94	79	94	91	100	82	90
Settlement	81	96	89	89	89	89	88	82	80	39
Forest	85	82	85	87	85	87	87	82	84	84
Wetland	100	98	98	98	98	98	84	88	100	87
OA (%)	82		84		85		87		89	

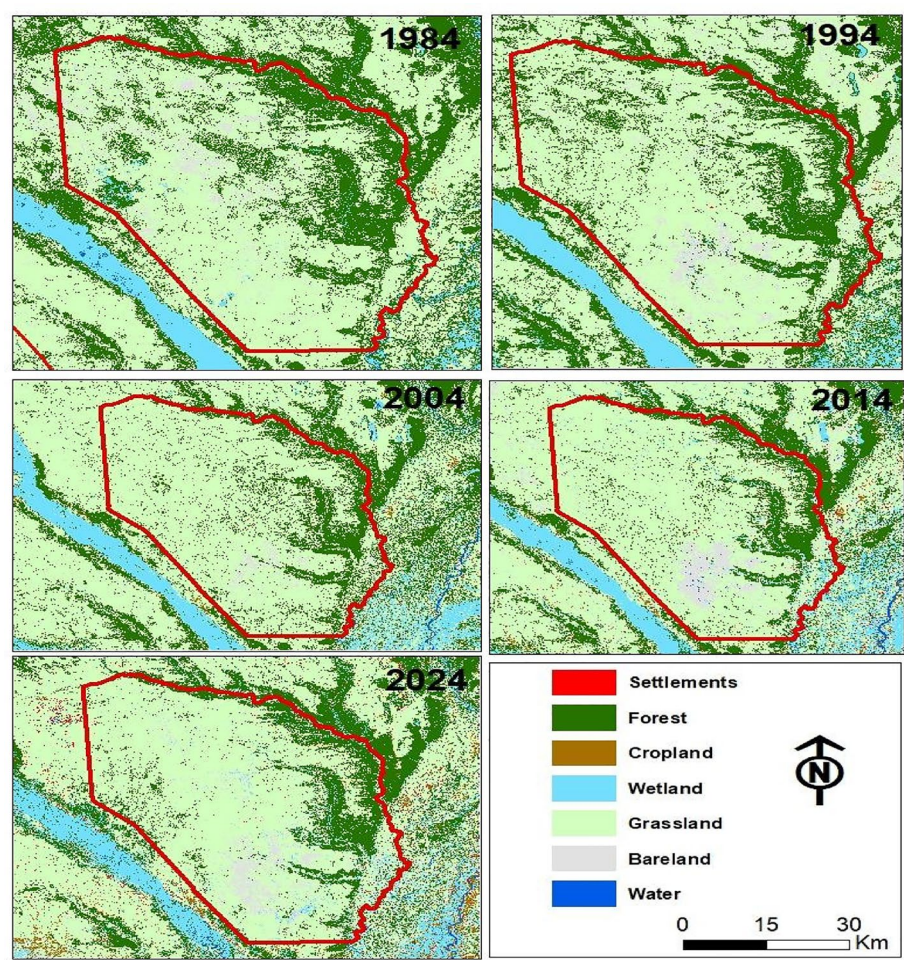


Fig. 3 Land cover maps for the Liuwa ecosystem between 1984 and 2024. This map shows the trends of the wild-life habitat in Liuwa ecosystem, both inside and outside the Liuwa Plain NP, western Zambia

involving settlements and bareland. The classification accuracy improvement ranged between 10 and 15% before (Table 3) and after (Table 4) the object-based refinement.

Figure 3 shows a classified map of the study area for each of the five periods and indicates that the Liuwa ecosystem have declined both grassland and forest cover over the study period. However, Liuwa ecosystem is predominantly covered by grassland, which

occupies over 75% of the total area. Furthermore, between 1994 and 2014, notable land cover changes occurred across the Liuwa ecosystem, with decreases in natural land cover types particularly forest and grassland and concurrent expansions in built-up areas and cropland. These changes are spatially concentrated in the southern part of the park, where development activity appears to be most intense, while the northern region remains relatively less affected. A visual inspection of Fig. 3 further highlights increases in non – habitat land cover types, such as Bareland, cropland, and settlements, particularly evident in 2024.

3.2 Habitat dynamics

Between 1984 and 2004, grassland increased slightly by 0.83% but subsequently declined by 0.65% between 2014 and 2024 (Fig. 4). In contrast to the fluctuating trend observed in grassland, forest cover exhibited a consistent decline throughout the study period, decreasing by up to 15%, apart from a brief stabilisation around 2014. Human-influenced land cover types, such as cropland and settlements, showed a continuous upward trend, with their combined spatial extent doubling from 1984 to 2024. Notably, wetlands and water bodies experienced a marked reduction in 2024, potentially because of the drought conditions reported during that year.

Furthermore, in comparing the extent of different land cover types in Liuwa Plain NP, grassland increased slightly by 0.83% between 1984 and 2004, but subsequently declined by 0.65% between 2014 and 2024. While both regions were dominated by grassland followed by forests throughout the study period, the extent and nature of change differed markedly. Inside the park (Fig. 5A), grassland increased by up to 5%, whereas outside the park (Fig. 5B), the increase was minimal, reaching only about 1%. Forest cover declined significantly outside the park, with a loss of up to 20%, compared to a more moderate 5% decline within the park. Furthermore, cropland and settlements expanded to more than three times their original extent outside the national parks, substantially greater than the changes observed within the protected area. These differences suggest that Liuwa National Park has played a critical role in mitigating human-induced land cover change, thereby contributing positively to habitat preservation within the park.

The marginal 1% grassland increase outside the park likely represents a transient degradation signal cleared or degraded woodland spectrally transitioning to grassland en

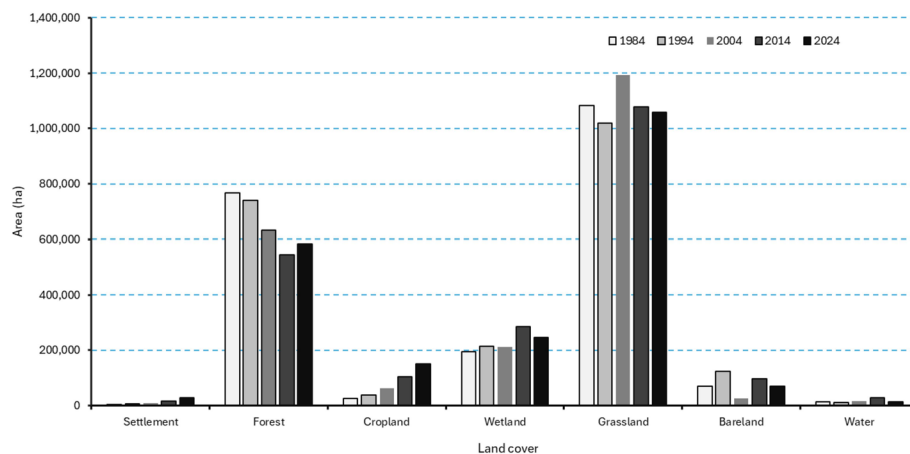


Fig. 4 Extent of land cover types including key habitats, such as grasslands, forests and wetlands in Liuwa Plain NP, western Zambia, between 1984 and 2024

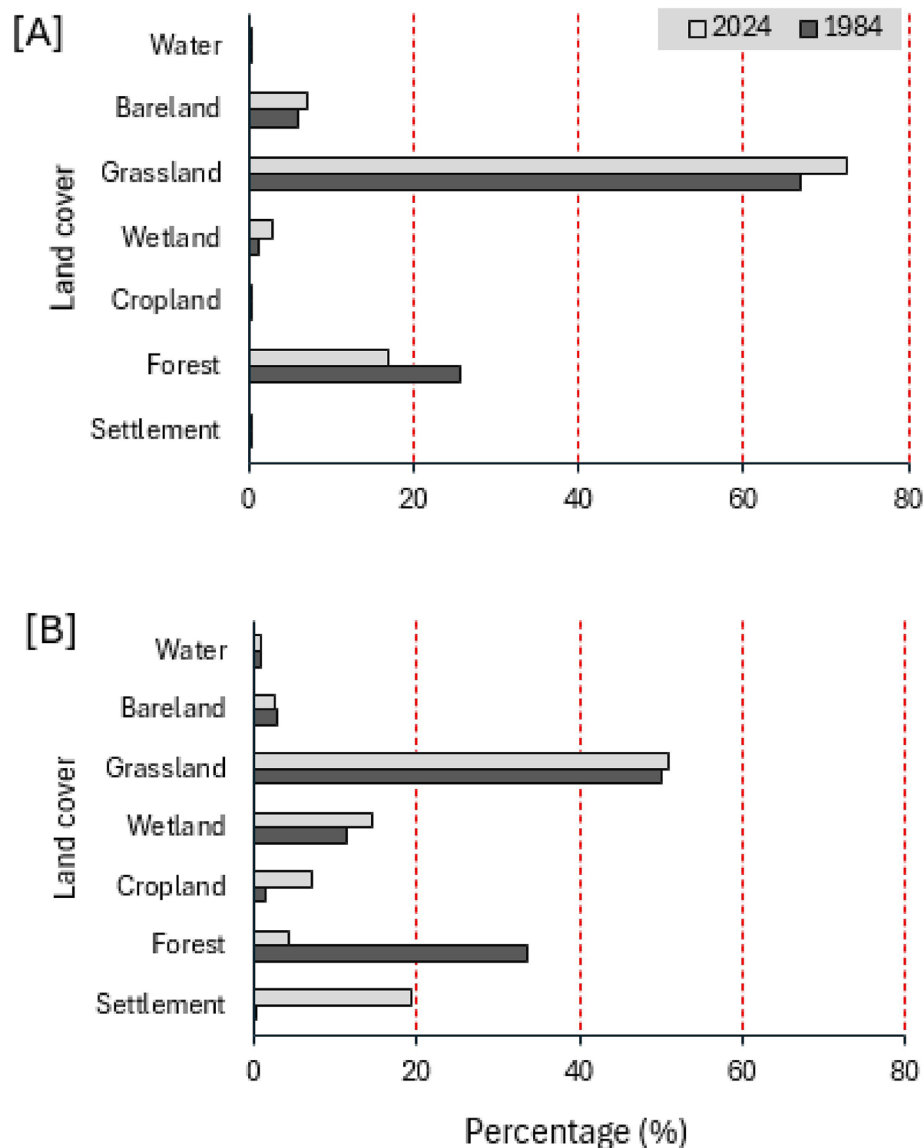


Fig. 5 Extent of land cover types including key habitats, such as grasslands, forests and wetlands with (a) inside and (b) outside the Liuwa Plain NP, western Zambia, between 1984 and 2024

route to cropland or bareland rather than genuine habitat recovery. This is supported by the concurrent 20% forest loss and the transition matrix, which identifies forest as the primary donor to grassland. Livestock grazing pressure with a well-documented driver in communal lands adjacent to protected areas across sub-Saharan Africa that could likely suppresses woody regeneration through selective browsing, thereby maintaining or slightly expanding open grassland cover [19]. Deliberate burning of bushland by local communities to create and maintain pastureland, a common practice in the Western Province of Zambia, may further contribute to this expansion, as dry season fires are widely recognised as agents of woodland-to-grassland conversion in miombo ecosystems [20]. Additionally, degraded or abandoned cropland reverting to secondary grassland over parts of the study period cannot be excluded as a contributing factor. Disentangling these drivers would require field-based validation and higher spatial resolution temporal analysis, and is therefore recommended as a priority for future research

as highlighted in conclusion of this paper. Besides, a general comparison of land cover changes inside and outside Liuwa National Park reveals a clear divergence in trends between the two areas.

3.3 Habitat loss and gain

Figure 6 shows the spatial distribution of major habitat dynamics within and surrounding Liuwa Plain NP. The most significant habitat losses were concentrated in the southern portion of the park, adjacent to Kalabo town. This area has experienced considerable expansion of built-up areas, agricultural land, and infrastructure developments, such as roads, all of which are contributing to habitat degradation and fragmentation. In contrast, the northern part of the park shows signs of persistent regrowth in both grassland and forest cover, suggesting a greater potential for ecological sustainability in that region.

A contextual analysis of land cover gains and losses within and outside Liuwa Plain NP reveals dynamic changes across all land cover types over the study period. Figure 7 indicates that both grassland and wetland experienced net gains, with gains exceeding losses in both areas. However, as visible in Fig. 7B, grassland gains and losses outside the park are approximately equal, suggesting a near-zero net change that reflects a dynamic equilibrium of grassland conversion and recovery rather than a directional trend. Although some forest regeneration was observed, particularly outside the park, forest losses especially outside the park were substantially greater than the gains. Notably, land cover types associated with habitat degradation, such as cropland and settlements, expanded significantly, with increases exceeding 300%, highlighting intense human pressure on the ecosystem.

4 Habitat transition

Table 5 presents habitat transitions by identifying the sources of gains and the destinations of losses for each land cover type. Both inside and outside Liuwa Plain NP, over 60% of the losses from habitat classes (forest, grassland, and wetland) were converted to cropland and settlements. Overall, grassland and forest areas were the primary contributors (donor classes), while cropland, settlements, and bareland were the main recipients (beneficiary classes). Although changes in wetlands and water bodies were relatively

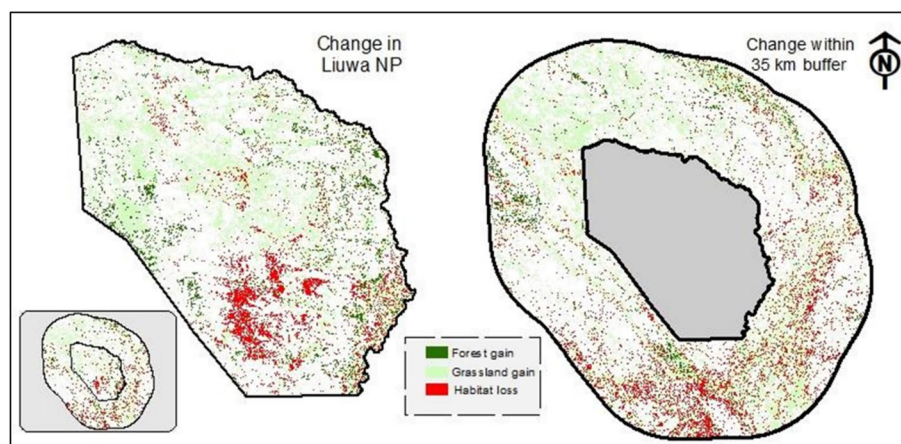


Fig. 6 Habitat losses within and outside Liuwa Plain NP, western Zambia, between 1984 and 2024. In this study habitat is defined as forest, grassland, and wetland

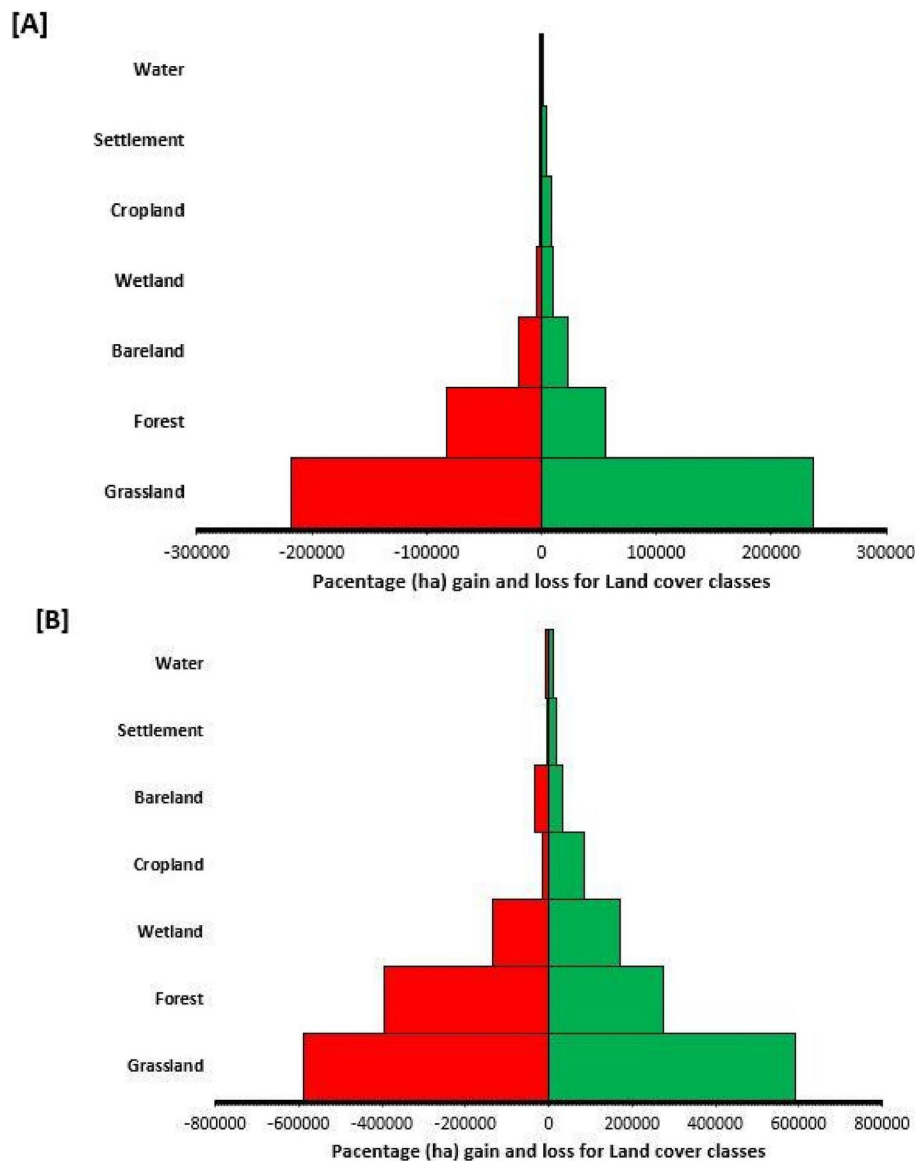


Fig. 7 Land cover gains and losses (km²) for **A** inside Liuwa Plain NP and **B** the 40-km buffer zone outside the park, calculated over the full study period (1984–2024)

minor, most of the observed transitions between these classes involved multiple classes; for example, wetlands transitioned not only to water but also substantially to grassland and forest, reflecting the spectral overlap acknowledged in the methods.

5 Discussion

The results of this study provide empirical evidence on the long-term habitat dynamics within the unique shared landscape in Liuwa ecosystem. The analysis is based on a series of high-accuracy thematic maps, each achieving classification accuracy exceeding 80% (Table 3). According to [18], such levels of accuracy are essential for reliable post-classification change detection and any further analysis. These classification accuracies (82–89%) are consistent with, and generally at the higher end of, those reported by comparable Landsat-based studies in ecologically similar environments in Western

Table 5 Habitat transition between 1984 and 2024 within (A) and outside (B) Liuwa Plain NP, western Zambia

To 2024									
	1984	Bareland	Cropland	Forest	Grassland	Settlement	Water	Wetland	Area Total
From (A)	Bareland	1,851.67	23.49	1,712.60	15,881.74	18.90	1.04	293.13	19,782.58
	Cropland	0.16	1.74		0.01				1.92
	Forest	1,974.98	275.05	39,179.75	36,770.10	111.56	75.89	4,999.05	83,386.38
	Grassland	18,716.61	882.35	13,336.83	181,642.23	264.59	5.98	3,484.28	218,332.86
	Settlement	0.05			0.10	0.86			1.00
	Water	7.15	0.33	20.71	12.39	0.00	59.09	54.98	154.65
	Wetland	243.19	15.84	1,017.26	2,134.77	6.27	44.20	716.87	4,178.40
	Area Total	22,793.80	1,198.79	55,267.15	236,441.35	402.17	186.22	9,548.31	325,837.79
To 2024									
	1984	Bareland	Cropland	Forest	Grassland	Settlement	Water	Wetland	Area Total
From (B)	Bareland	1,477.9	3,187.88	8,498.84	15,507.74	909.5	407.37	3,390	33,379.24
	Cropland	41.16	13,818	906.8	480.23	256.87	10.21	263.68	15,776.94
	Forest	7,487.2	20,064.06	173,636.36	136,157.7	3,830.83	940.02	54,071.41	396,187.55
	Grassland	20,030.24	42,385.32	68,551.07	424,850.03	11,640.03	195.57	21,894.24	589,546.5
	Settlement	78.34	667.24	88.22	246.04	890.52	8.89	114.03	2,093.28
	Water	296.12	525.4	383.92	162.73	129.63	4,241.98	3,726.03	9,465.8
	Wetland	2,517.21	3,913.81	23,928.37	13,647.32	964.24	3,226.8	85,904.33	134,102.07
	Area Total	31,928.16	84,561.71	275,993.58	591,051.76	18,621.61	9,030.83	169,363.72	1,180,551.38

Zambia, including [35] who achieved 81% in the adjacent Barotse Floodplain [36] who reported accuracies in the low-to-mid 80% range for the same floodplain system, [31] who attained 79–86% across Zambian woodlands using an object-based approach, and [37] who recorded 83–91% for Zambian forest reserves collectively indicating that the present study's classifications perform favorably within the range documented for spectrally complex, seasonally flooded landscapes in the region. The accuracies achieved in this study are generally consistent with other studies involving Landsat image classifications [32, 33] with Landsat 8 OLI/TIRS producing the highest accuracy. Moreover, the higher classification accuracy achieved with Landsat 8 OLI/TIRS is likely attributable to its improved radiometric resolution, enhanced signal-to-noise ratio, and superior spectral sensitivity compared to Landsat 5 TM, which improved the separability of spectrally similar land cover classes such as wetlands, grasslands, and bareland. Poursanidis et al. [38] indicates that the high accuracy of the OLI/TIRS image is likely due to the high quality of the Landsat 8 sensor. Given overall accuracies of 82–89%, some observed changes, particularly those of small magnitude, may fall within the bounds of classification uncertainty. The dominant trends reported here, forest decline, cropland and

settlement expansion, are substantially higher than the estimated classification uncertainty implied by the error matrices, providing confidence that these represent genuine landscape changes rather than classification artefacts.

The findings reveal a notable decline in land cover types associated with habitat integrity particularly outside the Liuwa Plain NP where forest cover declined by up to 20% and grassland by approximately 5% between 1984 and 2024. Besides, these results are consistent with a study in the Barotse Floodplain, the Zambezi river Sub Basin of Zambia which assessed and reported a general decline of the ecosystem services by [36] and also in line with a recent study in Malawi's Kasungu National Park [1] where substantial loss of forest cover, vegetation fragmentation and increased aggregation of remnant habitat between 1997 and 2018 were observed and these patterns were attributed to anthropogenic pressures and climate variability. In addition, similar habitat decline trends have been documented in other co-managed protected areas across Southern Africa, such as Zimbabwe's Gonarezhou National Park, where human-wildlife co-existence has led to complex land-cover transformations driven by settlement expansion and high elephant populations as reported by [39]. Additionally, in the same vein, a recent study in Okavango Delta of Botswana, revealed strong links between population growth and land cover changes, where rapid urban expansion from 0.68 km² to 9.61 km² between 1990 and 2020 corresponded with a marked decline in shrub and tree cover from 13.53 km² to 7.06 km², alongside notable water and vegetation fluctuations [40]. However, the interpretation of habitat dynamics in Liuwa Plain NP must also account for fire regimes, floodplain hydrology, and migration corridors, which act as confounding ecological drivers. Seasonal burning, annual flooding of the Liuwa ecosystem, and long-term herbivore movements continuously reshape vegetation structure and spectral reflectance patterns. The decline in natural land cover could also be partly influenced by livestock and wildlife grazing pressure, particularly during the late dry season when imagery was acquired and vegetation is at its minimum phenological cover.

Over the same period, human-induced land cover types, such as cropland and settlements, expanded more than threefold across the Liuwa ecosystem, highlighting growing anthropogenic pressure. Wetlands and water bodies experienced a relatively modest decline of less than 3%, largely attributed to the severe drought conditions that affected Zambia in 2023/2024 [35, 36]. The study also revealed that the water class decreased outside the park, while the water class within the park and the wetland class both within and outside the park showed net increases over the study period. Furthermore, the apparent mid-period year (2004) increases in wetland area both inside and outside the park are attributable to the natural hydrological dynamics of the Zambezi floodplain system, where inter-annual variability in flood pulse magnitude and rainfall seasonality causes wetland extent to fluctuate considerably between years. Since imagery was acquired at 10-year intervals, each classification snapshot reflects the hydrological condition of that specific year, a wet year naturally yields a larger mapped wetland extent than a dry year, independently of any structural long-term trend. This temporal sampling effect, rather than genuine permanent wetland expansion, most plausibly explains the observed intermittent increases. The overall 40-year trajectory nonetheless shows a net wetland decline, most pronounced in 2024 and robustly attributed to the exceptional 2023/2024 drought as corroborated by Climate Hazards Group InfraRed Precipitation with Station data (CHIRPS)-derived Standardised Precipitation Index (SPI) anomalies

and independently observed reservoir shrinkage [38, 41]. Future research should integrate SAR-based flood inundation mapping alongside optical classifications to better distinguish structural wetland loss from climatically driven fluctuations.

This study demonstrates that the habitat within and around Liuwa Plain NP has undergone noticeable changes. Although the overall margins of change remain minimal, suggesting that ecological integration is still largely intact, the ecosystem is under increasing pressure. Over the past decades, the human population that depends on and interacts with these natural resources has more than doubled, intensifying pressure on land and other ecosystem services. According to demographic data from the Zambia Statistical Agency, the population of Kalabo District increased from 89,689 in 1990 to approximately 120,000 in 2022 [42] representing a 34% increase that temporally corresponds with the observed expansion of cropland and settlements in the buffer zone, therefore, the northern park regrowth in grassland and forest observed and indicated in the results section of this study is most plausibly driven by its geographic remoteness from the southward concentrated human population pressure around Kalabo town, rather than active conservation intervention by tripitite governance structure by the African Parks Network with DNPW and the the BRE within the Liuwa Plain NP. These trend are consistent with the trend reported by [40] reported in Okavango, Delta of Botswana in a similar recent study. This population growth has fueled demand for land for settlement, agriculture, and fishing consistent with urban expansion trends observed elsewhere. For instance a geospatial analysis of Lucknow city, India, by [21] revealed fivefold built up area from 53.86 km² in 1991 to 261.45 km² in 2021, driven by rapid population growth and socio economic development. In the same vein the study aligned to Kolkata which has threefold between 1991 and 2018 [43] and Bengaluru with a fourfold growth in between 1992–2016 [44] in India as well as consistent with the results revealed in a study by [22] in Afghanistan. Similarly, in Liuwa Plain NP, population pressure has emerged as a key driver of land use transformations, highlighting the shared dynamics between rural and urban landscapes undergoing human induced spatial expansion in this rare case of human-wildlife co-existence landscape. In such shared ecosystems, the risk and frequency of human-wildlife conflict tend to rise significantly, requiring pragmatic countermeasures [42, 45, 46] such as the establishment of wildlife-proof fencing and buffer zones around community fields, community-based natural resource management (CBNRM) programmes that provide local communities with tangible conservation incentives, compensation schemes for livestock losses caused by predators, early warning systems for wildlife movement near settlements, and the promotion of alternative livelihoods such as ecotourism and sustainable aquaculture to reduce direct resource competition.

Beyond the fixed administrative park boundary used in this analysis, it is acknowledged that wildlife space use in Liuwa Plain NP is inherently dynamic and strongly mediated by seasonal hydrological conditions. The Liuwa floodplain undergoes dramatic seasonal inundation driven by the annual Zambezi flood pulse, fundamentally altering landscape accessibility for both wildlife and human communities between wet and dry seasons. During the dry season, wildlife concentrates within the core park area around permanent water sources, whereas wet season inundation expands accessible habitat and dispersal corridors well beyond the gazetted boundary, a pattern closely analogous to that documented by [47] in Amboseli National Park, Kenya, where wildlife ranges

contract during dry periods and expand into surrounding communal lands during wet seasons in response to freshwater availability. Incorporating dynamic, season-specific boundaries, delineated using GPS telemetry data or species distribution modelling would provide a more ecologically meaningful characterisation of functional habitat availability and human-wildlife spatial overlap across seasons. We fully recognise this as a valuable methodological advancement; however, its implementation in the current study was constrained by the absence of multi-temporal wildlife telemetry data and the 10-year classification interval, which precludes seasonal disaggregation. The fixed boundary analysis nonetheless provides a robust baseline against which future season-specific assessments can be benchmarked, and the observed divergence in land cover trends between inside and outside the park remains a valid and informative indicator of differential habitat protection effectiveness.

Our results reveal a notable impact of the 2023/2024 drought on the Liuwa landscape, with a significant decline in the extent of wetlands and water resources observed across the study area. Further, these results resonates with a study within Liuwa Plain NP by [48] who reported through community perspectives that droughts have been observed as a cause of decline in water levels in wetlands, contributing to some of the observed negative impacts on fish abundance and crop yield declines within the ecosystem. The study further, resonates with [49] who reported the decline of larger mamal populations in their study such as elephants and other key large herbivores and habitat loss in the Amboseli ecosystem of Kenya is in relation to droughts. Moreover, our interpretation that the 2023/2023 precipitated observable decline of wetlands and water in the Liuwa Plain NP and the surrounding areas is consistent with national scale diagnostics for the same season.

Using Climate Hazards Group InfraRed Precipitation with Station data (CHIRPS) v2 rainfall (1984 – March 2024) and Standardised Precipitation Index (SPI), computed at 1-,3- and 6- month windows, Chisanga et al. [45] report statistically significant trends indicative of short-term water deficit across Zambia and characterise seasons as increasingly dry, particularly at shorter accumulation periods relevant to hydrologic stress (ONDJFM) [45]. The same study documents CHIRPS source and availability (0.05° resolution, Jan 1981–Mar 2024). Complementing these meteorological indicators, Ghosh et al. [46] showed concurrent hydrological impacts between December 2023 and April 2024, Zambia experienced significant surface water losses in major reservoirs, e.g., Itezhi-Tezhi and Lake Kariba, reflecting basin-wide water deficits during the same period [46]. In this regard, taken together, nationwide SPI anomalies from CHIRPS and independently observed reservoir shrinkage during the 2023/2024 provide convergent, quantitative support for attributing the observed wetland and water declines in Liuwa Plain NP and the surrounding areas to the exceptional drought conditions of that season.

Despite the drought's severity, its effects did not immediately translate to wildlife mortality, suggesting a complex behavioral plasticity relationship between climate-driven resource scarcity and wildlife population dynamics. However, wildlife populations tend to respond to the environmental stressors, such as effects of climate change, variedly [39, 46]. The drought's consequences, including reduced water sources, diminished food quality and quantity, potential limitations in breeding habitats, and substantial declines in fish stocks, a crucial resource for local communities underscore the increasing vulnerability of both wildlife and human populations within this ecosystem. The remote

location of the park may further exacerbate these vulnerabilities. These findings are consistent with existing literature, such as [46, 50], which highlights the role of escalating drought frequency and severity in exacerbating human-wildlife conflicts in Southern Africa, as observed in Zimbabwe. This emphasises the urgent need for concerted efforts to enhance adaptation measures, thereby increasing the resilience of both wildlife and human populations to the impacts of climate change. By reducing vulnerability through targeted interventions, it is possible to mitigate the adverse effects of climate-driven stressors on ecosystems and communities [42, 43].

Although the management of Liuwa Plain NP has evolved over time, the habitat quality for wildlife remains sufficiently intact to support a rich and diverse ecosystem. While this study did not assess changes in wildlife populations directly, once severely depleted of wildlife, the park has since experienced a significant recovery [6]. The severe historical depletion of wildlife in Liuwa Plain NP was primarily attributed to a combination of factors, most notably rampant illegal hunting and poaching and reduced law enforcement capacity in the past few decades, which decimated large mammal populations, including wildebeest, buffalo, and lion. This was compounded by habitat degradation, reduced prey availability, and the collapse of traditional community-based wildlife governance structures. Disease outbreaks and periodic drought-induced freshwater scarcity further stressed already diminished populations, while the absence of effective management oversight allowed these pressures to operate unchecked over several decades. Besides, since 2003, Liuwa has been managed through a formal co-management arrangement involving three key stakeholders: the Department of National Parks and Wildlife (DNPW), the Barotse Royal Establishment (BRE), and African Parks Network (APN). This collaborative governance model emphasises conservation, law enforcement, land use planning, and the sustainable utilisation of natural resources. The current healthy state of the park's ecosystems stands as a testament to the effectiveness of this partnership and the long-term benefits of integrated management strategies and this is supported by the findings in a similar study conducted in Zimbabwe by [39] who strongly recommended and highlighted the importance of innovative partnerships between state and non-state profit organisations, local communities and local international partners in managing protected areas in Zimbabwe and elsewhere as they provide financial and technical assistance crucial in protected area administration and resource governance.

The observed decline in forests and grasslands, the land cover types most closely associated with wildlife habitat integrity has direct and well-established implications for wildlife populations in the Liuwa ecosystem. Forest loss reduces canopy cover and structural complexity that many species depend on for shelter, breeding, and thermoregulation, while grassland degradation diminishes the forage base that sustains the large herbivore populations, most notably the wildebeest migration, for which Liuwa is internationally recognised and being number two in Africa for Wildebeest migration from Tanzanian Serengeti. The land cover change results are broadly consistent with expectations given the known historical depletion and ongoing recovery trajectory of wildlife in the park habitat loss outside the boundary, where anthropogenic pressures are greatest, is likely to constrain wildlife dispersal and reduce the carrying capacity of the broader landscape. Conversely, the relative stability of habitat inside the park, supported by active co-management, provides a credible ecological explanation for the observed wildlife recovery since 2003. These findings therefore reinforce the conclusion that

maintaining habitat integrity within the gazetted boundary is a prerequisite for sustaining viable wildlife populations in this shared landscape.

The findings from this study make three key contributions to the management of Liuwa Plain NP: (1) they provide essential data to inform land use planning and park management; (2) they highlight the impact of the 2023/2024 drought, underscoring the need for integrated climate adaptation strategies for both wildlife and human communities; and (3) they emphasise the importance of strengthening conservation efforts to enhance habitat quality and support a wide range of biodiversity as similarly noted and indicated by [51] in their study [52] have previously demonstrated the value of geospatial information in wildlife habitat evaluation and planning. Building on that foundation, this study offers spatial insights that can be used to identify areas of persistent habitat loss, enabling more targeted zonation and the development of adaptive management strategies. These insights are critical to supporting long-term human-wildlife co-existence in Liuwa ecosystem. Furthermore, this study highlights practical examples of more effective conservation planning and habitat management strategies that could be applicable to Liuwa Plain NP and similar protected areas which include: (i) spatially targeted afforestation and natural regeneration programmes in identified deforestation hotspots along the park boundary, using remote sensing-derived priority maps to guide replanting efforts; (ii) establishment and enforcement of clearly demarcated buffer zones that separate community agricultural land from core wildlife habitat, reducing encroachment pressure; (iii) corridor restoration between fragmented habitat patches to facilitate wildlife movement and maintain genetic connectivity across the landscape; (iv) integration of community-based fire management protocols to regulate seasonal burning and prevent uncontrolled fires that degrade grassland and woodland structure; and (v) adoption of adaptive land use zoning frameworks informed by periodic satellite-based monitoring, enabling managers to reallocate conservation effort dynamically in response to detected habitat change. These strategies, when implemented within the existing co-management framework involving DNPW, BRE, and APN, have the potential to substantially reduce habitat loss and enhance ecosystem resilience in Liuwa and comparable human-wildlife co-existence landscapes globally.

The findings of this study should be interpreted with consideration of several limitations. First, although the land cover classifications achieved high accuracy, they were not error-free, and misclassifications are still possible; confidence intervals following [18] would further strengthen the reported estimates. Consequently, some land cover transitions may have been missed, or inaccurately estimated, either over or under-represented during the post-classification analysis. Second, testing data collection relied on visual interpretation against Google Earth imagery with limited independent field GPS ground-truth, which may introduce interpreter bias, particularly for historical periods with sparse Google Earth coverage. Third, the classification was conducted at 10-year intervals, which limits the ability to detect short-term or seasonal changes, such as those caused by fire, droughts, or other episodic events; future research should therefore consider shorter temporal intervals to capture such dynamics more effectively. In addition, the potential influence of fire regimes, flood dynamics, and migration corridors, all highly relevant to the Liuwa ecosystem were not investigated in detail in this study, and their conclusions are not fully addressed here; future research should integrate climate data such as CHIRPS datasets alongside these ecological drivers to draw more conclusive

results, particularly with respect to drought, which is central to the findings of this study. Finally, this study did not investigate the underlying drivers of habitat dynamics; an in-depth understanding of these drivers is essential for informed conservation planning, and future studies should aim to identify and analyse the socio-economic, environmental, and institutional factors influencing habitat dynamics in the Liuwa ecosystem.

6 Conclusions

This study aimed to assess four decades of habitat dynamics in Liuwa Plain NP, a human-wildlife co-existence ecosystem of Western Zambia. The Liuwa ecosystem uniquely supports the co-occurrence of human and wildlife populations, creating significant potential for competition over space and resources. The results revealed a general decline in key land cover types closely associated with wildlife habitats, particularly forests and grasslands. While similar trends were observed both inside and outside the park, habitat loss was significantly more pronounced in areas outside the park boundaries. Human-induced land uses, including cropland and settlements, showed a marked increase in spatial extent over the study period. Although wetlands and water bodies did not change remarkably over the period, a noticeable decline was observed in 2024, likely due to the 2023/2024 drought and this is in line with nationwide SPI anomalies from CHIRPS and independently observed reservoir shrinkage during the 2023/2024, which provides a convergent, quantitative support for attributing the observed wetland and water declines in Liuwa Plain NP and the surrounding areas to the exceptional drought conditions of that season. These findings provide valuable insights that can support more effective conservation planning and inform habitat management strategies aimed at minimising habitat loss. Therefore, while the study advocates for the eventual adoption of remote sensing-based monitoring systems capable of providing near-real-time data on habitat condition and change, the implementation of such systems remains a longer-term aspirational goal given the current technical, financial, and institutional constraints. In the interim, periodic satellite-based assessments at 5–10-year intervals, as demonstrated here, offer a practical and replicable monitoring framework. Such monitoring tools can guide targeted restoration interventions, for instance, identifying priority zones for afforestation, corridor establishment and natural regeneration to mitigate fragmentation and strengthen ecosystem resilience to climate change. Finally, the study recommends the integration of satellite-derived information into adaptive management frameworks, which will enable evidence-based decision making, ensuring that conservation actions in Liuwa Plain NP and similar protected areas across the globe remain responsive to spatio-temporal habitat trends. Future research should prioritise higher temporal resolution mapping using high spatial resolution imagery to better capture short-term changes and focus on identifying the underlying drivers of land cover change in the Liuwa ecosystem. Had high spatial resolution imagery been available across all five time periods, an object-based image analysis (OBIA) approach combined with Random Forest classification would have been considered, as this would improve delineation of spectrally mixed classes such as wetlands and seasonally flooded grasslands; however, for multi-decadal analysis, the Landsat archive remains the most consistent and cost-effective data source, and Random Forest remains an appropriate and robust classifier for this data type. Furthermore, future research should consider exploring alternative and advanced machine learning classification algorithms, such as Support Vector Machine

(SVM), and deep learning-based approaches, which have demonstrated superior performance in complex land cover mapping contexts and may improve classification accuracy and thematic detail beyond what was achievable with the approach employed in this study. While the implementation of near real-time remote sensing monitoring is advocated by scholars, several realistic challenges must be acknowledged. These include limited internet connectivity in several sub-Saharan countries, high costs of high spatial resolution imagery, insufficient technical capacity at institutional levels, lack of operational protocols for translating spatial data into management decisions, and persistent cloud cover that compromises optical imagery quality in tropical ecosystems. Overcoming these barriers will require multi-stakeholder partnerships and investment in open-access cloud computing platforms, such as Google Earth Engine, which can significantly lower technical and financial barriers to near real-time habitat monitoring in resource-constrained conservation settings.

Acknowledgements

We wish to thank Copperbelt University for providing support during the preparation of the manuscript.

Author contributions

DCM and GM analysed, DP and DCM drafted the manuscript, while NN, DZ and VRN conceptualised and reviewed the manuscript.

Funding

Funding was provided by Southern African Science Service Centre for Climate Change and Adaptive Land Management (SASSCAL) under the Enhanced Livelihoods and Natural Resource Management under accelerated Climate Change (ELNAC) project supported by the Federal Ministry of Research, Technology and Space of Germany.

Data availability

Data will be provided upon reasonable request to the corresponding author. The data is held in Google Drive and a link will be provided.

Declarations

Ethics approval and consent to participate

This study was conducted under the ethical clearance No.2023/2/13/1 issued by The Copperbelt University, and research permit No. NPW/8/27/1 issued by Department of National Parks and Wildlife (DNPW). Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Received: 5 January 2026 / Accepted: 11 July 2026

Published online: 17 July 2026

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